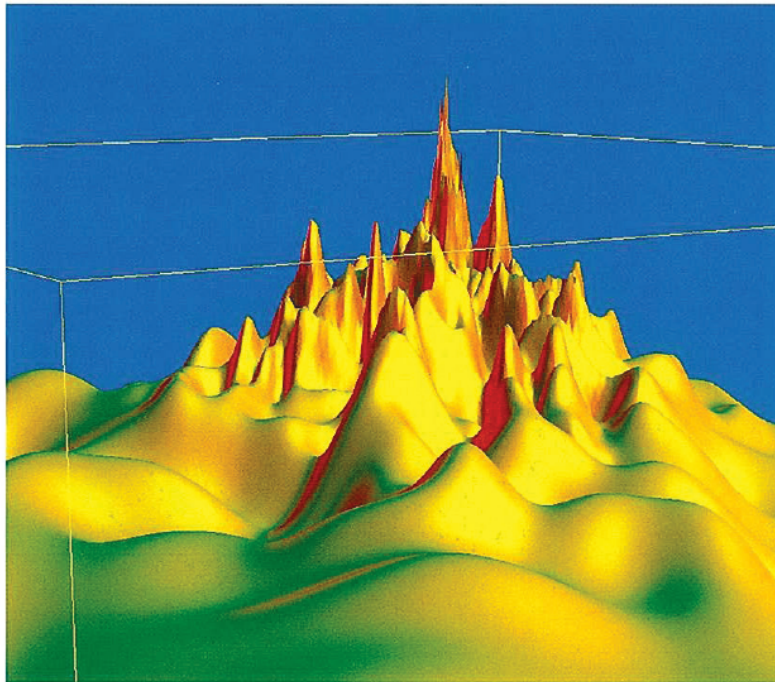


THIRD EDITION

STRATEGY
and the
BUSINESS
LANDSCAPE

PANKAJ GHEMAWAT



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CHAPTER

3

Creating Competitive Advantage

Pankaj Ghemawat and Jan W. Rivkin

*If a man . . . make a better mousetrap than his neighbor,
tho' he build his house in the woods, the world will make
a path to his door.*

—RALPH WALDO EMERSON (ATTRIBUTED)

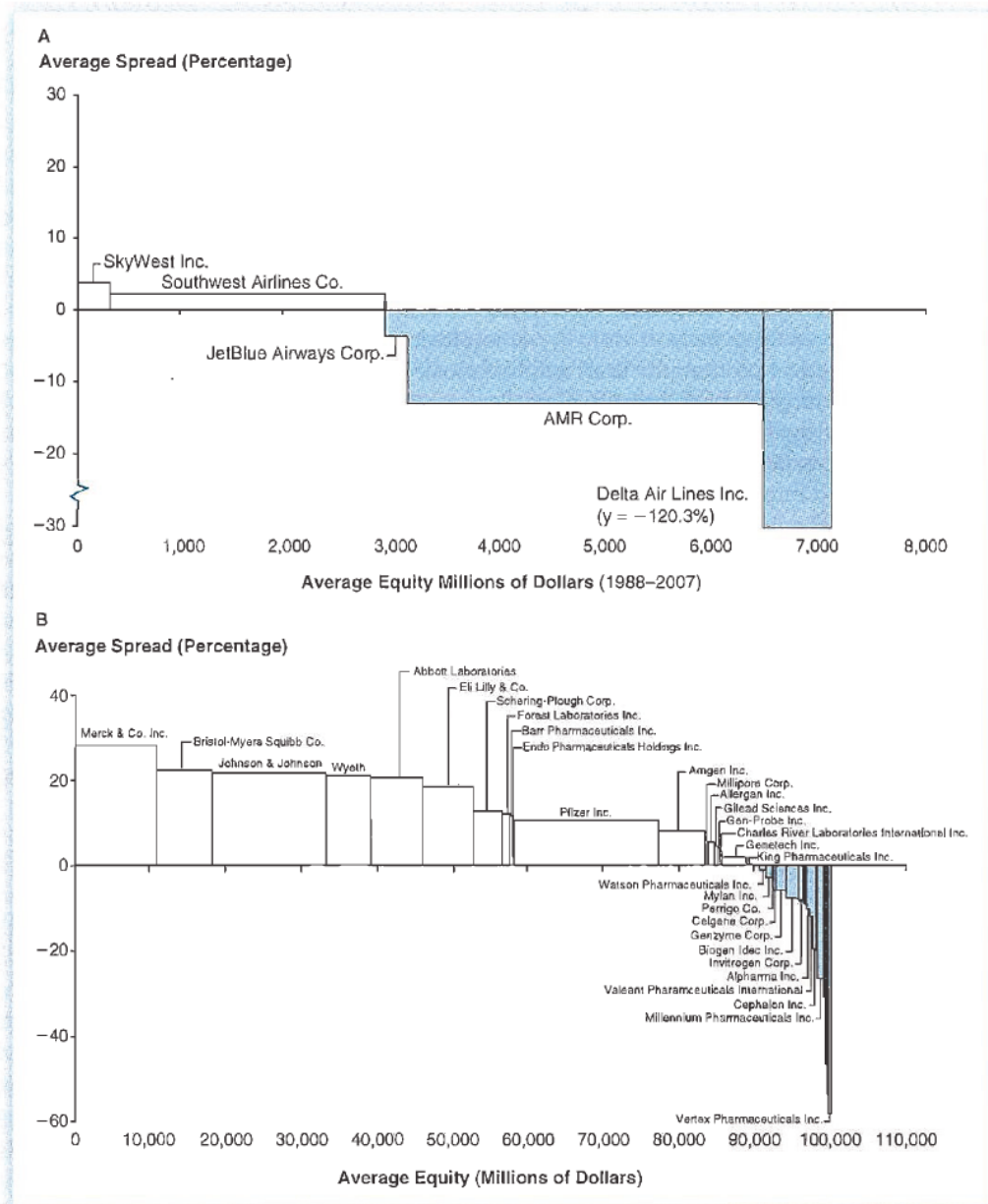
Lecturing in the nineteenth century, Emerson anticipated one of the key pieces of advice that strategists still stress at the beginning of the twenty-first: Strive for an edge over the competition! Since Emerson's time, strategists have explored, in effect, what makes one mousetrap better than others from the point of view of profitability. "Better" involves not just buyers' **willingness to pay** for a firm's mousetrap or the costs incurred to provide the trap, but the difference between the two. **In other words, an appreciation of the importance of a competitive edge has been elaborated into an understanding of the economics of what might be called the competitive wedge. A firm is said to have created a competitive advantage over its rivals if it has driven a wider wedge between willingness to pay and costs than its competitors have achieved.¹**

The long-standing interest in competitive advantage and its inclusion as the second dimension of the profitability grid in Exhibit 1.7 reflect the sense that while industry- or population-level effects have a large impact on business performance, large differences in performance also appear *within* industries. Consider, for example, the two industries identified as outliers in terms of performance and analyzed in Chapter 2—pharmaceuticals and airlines. Exhibit 3.1 unbundles the spreads between returns on equity and the costs of equity capital in these two industries, competitor by competitor.² Note that while pharmaceutical firms have, on average, earned returns higher than airlines, the best performing airlines create economic value while the worst performing pharmaceutical firms destroy it. In other words, while industry matters, it is not destiny.

EXHIBIT 3.1 A Average Economic Profits in the Airline Industry, 1988–2007.

(Source: © 2008 Marakon/Trinsum) **B Average Economic Profits in the Pharmaceutical Industry, 1988–2007.** (Source: © 2008 Marakon/Trinsum)

Note: Several familiar airlines such as United, Northwest, and USAir are invisible in Exhibit 3.1 because of negative book equity (bankruptcy). What this means, of course, is that the depiction in the exhibit understates actual value destruction in the airline industry.



The structure within industries, often described in terms of “strategic groups,” helps explain part of these differences in performance. Thus, while the major pharmaceutical firms in Exhibit 3.1 have averaged returns 20 percent higher than their cost of equity capital, biotechnology firms as a group have averaged returns 5 percent lower than their cost of capital. But there is more to the story than just group-level differences. Within the group of biotech companies, Amgen has historically outperformed its peers as has Southwest Airlines (although both have come under pressure recently, foreshadowing the discussions of the dynamics of competitive advantage in Chapters 4 and 5).

More systematic research confirms that such within-industry performance differences are widespread and substantial. The research suggests, in fact, that within-industry differences in profitability are larger than differences in the averages across industries. While industry-level effects appear to account for 10–20 percent of the variation in business profitability in the United States, stable within-industry effects account for 30–45 percent.³ (Most of the remainder can be assigned to effects that fluctuate from year to year.) Note, however, that this is *not* an argument for ignoring industry-level effects, the other dimension of the profitability grid introduced in Exhibit 1.7 and discussed in Chapter 2. First, industry-level effects *do* account for a significant fraction of the variation in performance observed on average. Second, industry-level effects may have a more persistent influence on business-level profitability than within-industry differences.⁴ Third, there is some evidence that industry characteristics play a larger role in determining the room for positive departures from average profitability than their average effects suggest.⁵ Finally, market leaders, in particular, often confront important tensions between managing industry structure and improving their own competitive position within that structure.

All that said, within-industry differences in profitability do appear to be larger than the differences in the averages across industries—and may loom even larger during downturns. In purely arithmetic terms, a deep downturn that pushes an industry’s average player into unprofitability makes a superior competitive position the only basis for profitability. And in terms of strategy, firms are often exhorted to invest during a downturn to improve their long-term competitive positions.⁶ Such advice is meant to counteract a tendency by businesses to cut investments during downturns more than can be explained by either changes in relevant (longer-run) economic forecasts or externally-imposed capital constraints. A number of mechanisms may underlie such a pattern, including

- manic depressive behavior of the sort evoked by Keynes in discussing animal spirits
- herd behavior by competitors that can lead to cascading cut-backs across them
- the managerial bias toward internal funding even when external funding is available on efficient terms, implying an internally-imposed capital constraint.
- a focus by managers during downturns on firm survival rather than maximization of expected shareholder value

In order to understand within-industry differences in good times or bad, we must zoom in from the industry level to look at the landscapes within industries. Examining intra-industry landscapes, which also turn out to be very rugged, is the focus of this chapter. The first part of the chapter reviews the historical development of the core concepts in analyzing competitive position (either advantaged or disadvantaged): competitive cost analysis, the analysis of differentiation, cost-benefit trade-offs, and added

value. The second part of the chapter draws on these concepts to lay out a process for analyzing competitive positioning, illustrated with an extended example. This tack, which is primarily analytical, is not intended to deny the importance of creativity and insight in the creation of competitive advantage. Rather, it can be read as an attempt to guide entrepreneurial energies by setting up a battery of tests for new business ideas. The topic of creativity in developing new ideas is taken up again in Chapter 4.

THE DEVELOPMENT OF CONCEPTS FOR COMPETITIVE POSITIONING

Starting in the 1970s, academic researchers made a number of contributions to our understanding of positioning within industries. The IO-based approach to strategic groups, initiated at the Harvard Business School by Michael S. Hunt's work on broad-line versus narrowline strategies in the major home appliance industry, suggested that competitors within particular industries could be grouped in terms of their competitive strategies in ways that helped explain their interactions and relative profitability.⁷ And a stream of work at Purdue University by Dan E. Schendel and his collaborators explored the heterogeneity of competitive positions, strategies, and performance in brewing and other industries with a combination of statistical analysis and qualitative case studies.⁸ The work that seems to have had the most impact on business-strategic thinking about competitive positions in the late 1970s and the 1980s, however, was more pragmatic than academic in its orientation. Consultants once again played a leading role, particularly in the development of techniques for competitive cost analysis.

Cost Analysis

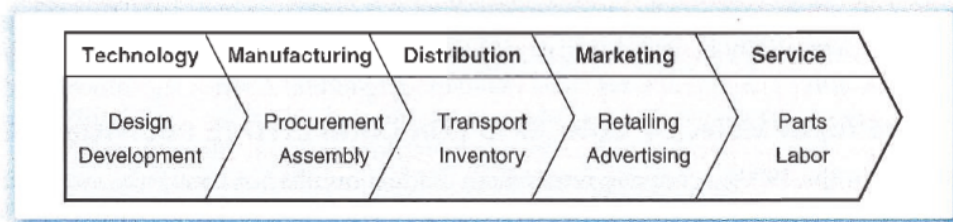
With the growing acceptance of the experience curve in the 1960s, most strategists turned to some type of cost analysis as the basis for assessing competitive positions. The interest in competitive cost analysis survived the declining popularity of the experience curve in the 1970s but was reshaped by it in two important ways. First, greater attention was paid to disaggregating businesses into their components as well as to assessing how costs in a particular activity might be shared across businesses. Second, strategists greatly enriched their menu of cost drivers, expanding it beyond just experience.

The disaggregation of businesses into components was motivated, in part, by early attempts to "fix" the experience curve so as to deal with the rising real prices of many raw materials in the 1970s.⁹ The proposed fix involved splitting costs into the costs of purchased materials and "cost added" (value added minus profit margins) and then redefining the experience curve as applying only to the latter category. The natural next step was to disaggregate a business's entire cost structure into parts—functions, processes, or activities—whose costs might be expected to behave in interestingly different ways. (To be consistent with later sections of the chapter, we will refer to the parts as "activities.")¹⁰ As in the case of portfolio analysis, the idea of splitting businesses into component activities diffused quickly among consultants and their clients in the 1970s. A template for activity analysis that became especially prominent is reproduced in Exhibit 3.2.

Activity-based analysis also suggested a way of circumventing the "free-standing" conception of individual businesses built into the concept of strategic business units (SBUs).¹¹ One persistent problem in splitting diversified corporations into SBUs was

EXHIBIT 3.2 McKinsey's Business System.¹²

(Source: Pankaj Ghemawat, "Competition and Business Strategy in Historical Perspective," *Business History Review*, 76 (Spring 2002).)



that, with the exception of pure conglomerates, SBUs often shared elements of their cost structure with one another. Consulting firms—particularly Bain & Company and Strategic Planning Associates (SPA)—began to emphasize the development of field maps or matrices that identified shared costs at the level of individual activities that were linked across businesses.¹³

In another important development in competitive cost analysis during the late 1970s and early 1980s, strategists began to consider a richer menu of cost drivers. Scale effects, although officially lumped into the experience curve, had long been studied independently in individual cases. Even more specific treatment of the effects of scale was now forced by activity analysis that might indicate, for example, that advertising costs were driven by national scale, whereas distribution costs were driven by local or regional scale. Field maps underscored the potential importance of economies (or diseconomies) of scope across businesses rather than scale within a given business. The effects of capacity utilization on costs were dramatized by macroeconomic downturns in the wake of the two oil shocks. The globalization of competition in many industries highlighted the location of activities as a key driver of competitors' cost positions and so on. Cost drivers are discussed more comprehensively in the second section of this chapter.

Differentiation Analysis

Increasingly sophisticated cost analysis was followed, after a significant lag, by **greater attention to customers in the process of analyzing competitive position**. Of course, customers had never been entirely invisible. Even in the heyday of experience curve analysis, market **segmentation** had been an essential strategic tool—but it was often used to chop up markets to "demonstrate" a positive link between segment share and cost advantage rather than for truly analytical purposes. By one insider's recollection (that of Walker Lewis, the founder of Strategic Planning Associates), "To those who defended the classic experience-curve strategy, about 80 percent of the businesses in the world were commodities."¹⁴ In the 1970s, this view began to change.

As strategists paid more attention to customer analysis, **they began to reconsider the idea that attaining low costs and offering customers low prices was always the best way to compete. Instead, they focused more closely on differentiated ways of competing that might let a business command a price premium by improving customers' performance** or reducing their (other) costs.¹⁵ Although (product) differentiation had

always occupied center stage in marketing, the idea of considering it in a cross-functional, competitive context alongside cost levels apparently began to emerge in business strategy in the 1970s. Thus one member of Harvard's Business Policy group assigned Joe Bain's writings on entry barriers (see Chapter 2) to students in the 1970s and used the concepts of cost and differentiation—implicit in two of Bain's three sources of entry barriers—to organize classroom discussions.¹⁶ McKinsey began to apply the distinction between cost and value in its consulting activities later in the same decade.¹⁷ The first extensive treatments of cost and differentiation, in Michael E. Porter's *Competitive Strategy* and in a *Harvard Business Review* article by William K. Hall, appeared in 1980.¹⁸

Porter's 1985 book, *Competitive Advantage*, suggested analyzing cost and differentiation via the **value chain**, a template that is reproduced in Exhibit 3.3. Although Porter's value chain bore some resemblance to McKinsey's business system, his discussion of this construct emphasized the importance of regrouping functions into the activities actually performed to produce, market, deliver, and support products, thinking about linkages among activities, and connecting the value chain to the determinants of competitive position in a specific way:

Competitive advantage cannot be understood by looking at a firm as a whole. It stems from the many discrete activities a firm performs in designing, producing, marketing, delivering, and supporting its product. Each of these activities can contribute to a firm's relative cost position and create a basis for differentiation. The value chain disaggregates a firm into its strategically relevant activities in order to understand the behavior of costs and the existing and potential sources of differentiation.¹⁹

EXHIBIT 3.3 Value Chain for an Internet Start-Up.

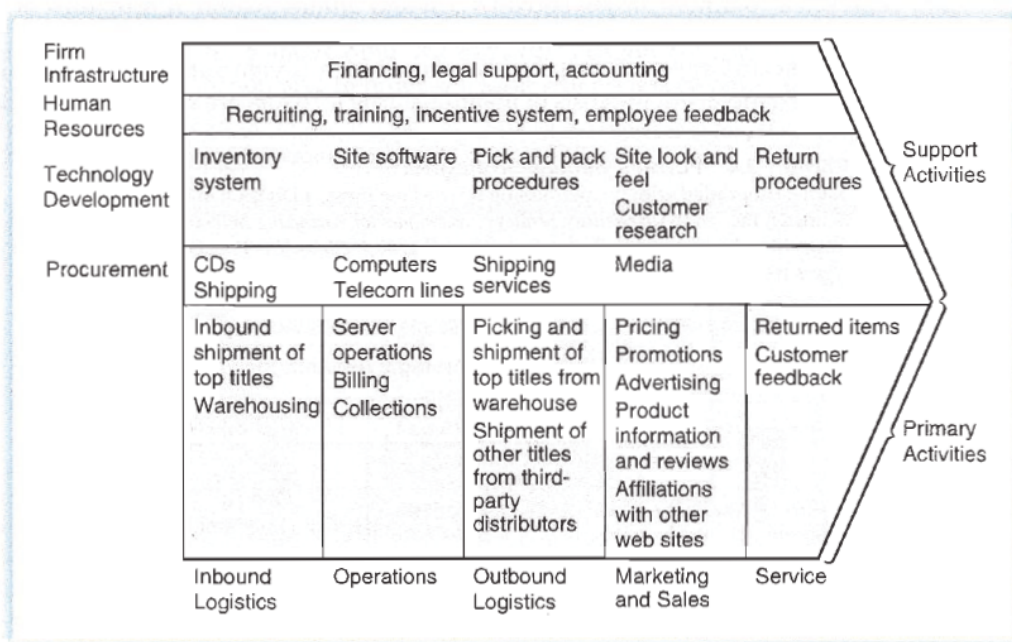


Exhibit 3.3 illustrates the value chain for an Internet start-up that sells and physically distributes music.

Subsequent advances in the integration of cost analysis and differentiation analysis not only disaggregated businesses into activities (or processes) but also split customers into segments based on cost-to-serve as well as customer needs. Such deaveraging of customers was often said to expose situations in which 20 percent of a business's customers accounted for more than 80 percent (or even 100 percent) of its profits.²⁰ It also suggested new customer segmentation criteria. Thus, starting in the late 1980s, Bain & Company built a thriving "customer retention" practice based on the notion that it generally cost more to capture new customers than to retain existing ones.

Costs Versus Differentiation

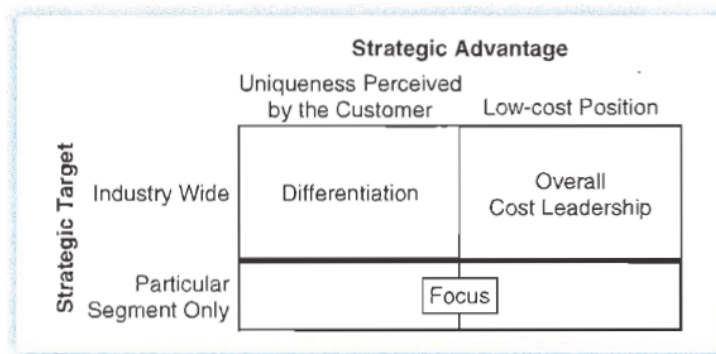
Porter and Hall, the first two strategists to write about both cost and differentiation, argued that successful companies usually had to choose to compete *either* on the basis of low costs *or* by differentiating products through quality and performance characteristics. Porter popularized this idea in terms of **the generic strategies** of low cost and differentiation. He also identified a **focus strategy** that cut across the two basic generic strategies (see Exhibit 3.4) and linked these strategy options to his work on industry analysis:

In some industries, there are no opportunities for focus or differentiation—it's solely a cost game—and this is true in a number of bulk commodities. In other industries, cost is relatively unimportant because of buyer and product characteristics.²¹

These generic strategies appealed to strategists for at least two reasons. First, they captured a common tension between cost and differentiation. A firm must often incur higher costs to deliver a product or service for which customers are willing to pay more. Most customers are willing to pay more for a Toyota automobile than for a Hyundai, for example, but the costs of manufacturing a Toyota are significantly higher

EXHIBIT 3.4 Porter's Generic Strategies.

Source: (Reprinted with the permission of The Free Press, a Division of Simon & Schuster, Inc., from *Competitive Strategy: Techniques for Analyzing Industries and Competitors* by Michael E. Porter. Copyright © 1980, 1998 by The Free Press. All rights reserved.)



than the costs of making a Hyundai. Toyota's higher profit margins derive from the fact that the price premium Toyota can command is greater than the incremental costs associated with its product.

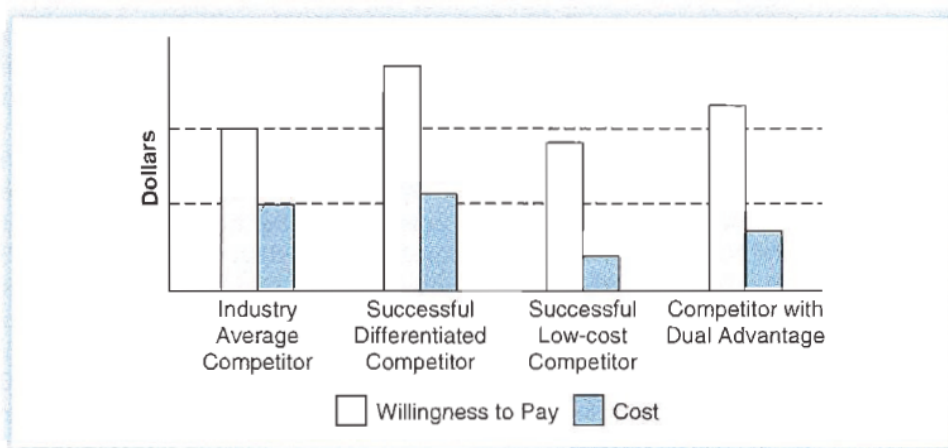
Second, the generic strategies were also appealing because the capabilities, organizational structure, reward system, corporate culture, and leadership style needed to make a low-cost strategy succeed are, at first sight, contrary to those required for differentiation. **For the sake of internal consistency and to maintain a single-minded purpose, a firm might have to choose to compete either in one way or the other.**

The generic strategies nonetheless provoked a vigorous debate among strategists, for both empirical and logical reasons. Empirically, the tension between cost and differentiation does not appear to be absolute. In the 1970s and 1980s, for instance, Japanese manufacturers in a number of industries found that, by reducing defect rates, they could make high-quality products at lower cost. Exhibit 3.5 traces the interplay between cost and differentiation in an expanded treatment of competitive advantage that **recognizes the possibility of dual competitive advantages.**

How common are companies with dual competitive advantages? **Porter argues that dual advantages are rare,** typically being based on operational differences across firms that are easily copied.²² Others contend that rejecting the trade-offs between cost and differentiation—replacing trade-offs with “trade-ons”—represents a fundamental way to transform competition in an industry.²³ The debate continues today.

A second challenge to the notion of generic strategies is logical in nature. **Although a desire for internal consistency may drive companies to the extremes of low cost and high differentiation, external considerations may pull firms back toward the center. If most customers want neither the simplest nor the most elaborate product, for instance, the most profitable strategy may be to offer a product of intermediate quality and intermediate costs.** Consider Zara, the Spanish fashion retailer. Its offerings are not nearly as upmarket as Gucci's or Armani's, although they sometimes share the same look. But Zara is a rung above other mass-market fashion retailers: its emphasis on within-season production and trendiness boosts willingness to pay—and production

EXHIBIT 3.5 Interplay between Cost and Differentiation.



costs as well. Fortunately for Zara, the incremental willingness to pay appears to exceed the incremental costs, making its intermediate position viable.

Strategists have generally stopped being dogmatic about generic strategies of any particular stripe. Instead, they embrace the idea that any analysis of competitive position must consider both relative cost and differentiation and recognize the tension between the two. Positioning, in this view, is or should be about driving the largest possible wedge between cost and differentiation. As differentiation rises, so too does cost in most instances. However, the largest gap between the two need not occur at the extremes of low costs or high price premia. The optimal position, from this perspective, represents a choice from a spectrum of trade-offs between cost and differentiation rather than a choice between mutually exclusive generic strategies.²⁴

A few examples will help illustrate the richness of positioning possibilities.

- Southwest Airlines has configured itself to focus on budget customers. It has standardized its fleet around fuel-efficient Boeing 737s, concentrates on short-haul point-to-point routes between midsize cities and secondary airports, offers very low ticket prices and no-frills service (no assigned seats, food service, baggage transfer, or connections with other airlines), emphasizes quick turnaround times, and manages to keep its planes in the air one-third longer each day than the average airline. Its stripped-down offering may generate slightly less willingness to pay than the offering of a full-service airline, but it incurs far lower costs than a full-service rival. As a result, Southwest is the only major U.S. airline to have been consistently profitable since deregulation (see Exhibit 3.1), while growing rapidly and maintaining one of the youngest fleets and the lowest debt levels among the majors.
- Accenture is regarded as the leader in information technology (IT) consulting because of its deep experience, reinforced by its research and development, its Knowledge Xchange portal, its stringent hiring and training practices, its close relationships with the CEOs as well as the CIOs of global corporations, and its successful attempts to build its brand. Some of the activities that Accenture undertakes to differentiate itself are clearly costly: R&D and training, for instance, swallow up 5 percent of revenues, even though they have been scaled back in recent years. On the other hand, Accenture often faces no competing bids when it pitches consulting work and is able to keep its revenues per consultant high, especially by maintaining a high utilization rate. This more than compensates for the extra costs it incurs: Accenture has historically earned returns significantly higher than most other large IT services companies.
- Cirque du Soleil is an innovative firm that combines elements of circus and theater. In designing its performances, Cirque excluded many of the high-cost components of the traditional circus—animals, star performers, and three-ring shows—and focused on what it considered to be the three elements responsible for the lasting allure of the circus: the clowns, the tent, and the acrobatic acts. By refining the clowns' acts, glamorizing the tent, and incorporating elements from the world of theater—themes and storylines, for example—Cirque de Soleil created a new category of entertainment with which it is synonymous.²⁵ In 2006, more than 7 million people paid a total of \$620 million to see its live performances, and the value of this privately held firm was estimated at \$1.5 billion.²⁶

Added Value

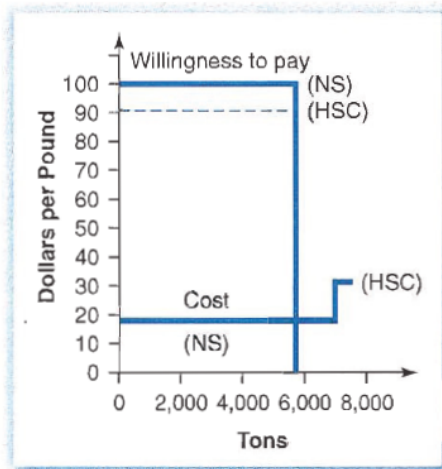
In the mid-1990s, Adam M. Brandenburger and Harborne W. Stuart added rigor to the idea of driving the largest possible wedge between costs and differentiation through their characterization of added value.²⁷ Brandenburger and Stuart considered three-stage vertical chains (suppliers → competitors → buyers) and were precise about the monetary quantities of interest. On the demand side, they mapped differentiation into buyer willingness to pay for products or services; on the supply side, they used the symmetric notion of supplier **opportunity costs** (the smallest amounts that suppliers would accept for the services and **resources** required to produce specific inputs). Given these definitions, the total **value** created by a transaction is the difference between the customer's willingness to pay and the supplier's opportunity cost. The division of this value among the three levels of the vertical chain is, in general, indeterminate. Nevertheless, an upper bound on the value captured by any player is provided by its added value—the maximum value that can be created by all the participants in the vertical chain minus the maximum value that would be created without that particular player.

More precisely, the amount of value that a firm can claim cannot exceed its added value under **unrestricted bargaining**. To see why this constraint applies, assume for a moment that a lucky firm *does* strike a deal that allows it to capture more than its added value. The value left over for the remaining participants is then less than the value that they could generate by arranging a deal among themselves. The remaining participants could, after all, break off and form a separate pact that improves their collective lot. Any deal that grants a firm more than its added value is vulnerable to such breakaway possibilities.

For an illustration of the usefulness of this style of analysis, consider the ill-starred decision in 1986 by the Holland Sweetener Company (HSC) to enter the aspartame industry, then monopolized by NutraSweet.²⁸ HSC's costs were probably higher, even after its initial capital investment was sunk, than those of NutraSweet because of limited scale and learning. In addition, customers' willingness to pay for HSC's aspartame was probably lower because of NutraSweet's heavy investments in building up its brand identity. HSC decided to enter the market anyway, presumably because the prices that Coca-Cola and PepsiCo were paying NutraSweet were approximately three times as high as HSC's prospective costs.

As things turned out, the big winners from HSC's entry were Coke and Pepsi, which were able to extract much lower prices from NutraSweet. An explanation is suggested in Exhibit 3.6, which graphs in stylized terms the willingness to pay and the relevant post-entry costs. Evidently, HSC's entry depressed NutraSweet's added value (hence the lower prices). HSC, however, could not expect to have added value post-entry (given unrestricted bargaining) because total value created would not be reduced if the firm were to disappear.²⁹ So HSC would have had to do something more than just jump into this business if it wanted to earn profits from it. One possibility would have been to convince Coke and Pepsi to pay HSC to play up front, instead of relying on their "goodwill" after its entry to amortize its fixed costs.³⁰ Another approach would have been to communicate that HSC's market-share goals were sufficiently modest that it made no sense for NutraSweet to lower its prices across the board.³¹

Added value is sometimes quantifiable, as was approximately the case in the context of aspartame. Even when it isn't, however, it provides a useful test for judging a

EXHIBIT 3.6 Costs and Willingness to Pay for Aspartame.

firm's strategy: If a firm were to disappear, would someone in its network of suppliers, customers, and complementors miss it? This question is harder edged than older tests in the same vein—come up with a better product (à la Emerson), manage for uniqueness, focus on your distinctive competence, and so on—because it is based on an explicit model of interactions among buyers, suppliers, competitors, and complementors.

The concept of added value also helps tie together intra-industry analysis of competitive advantage and industry-level analysis of average profitability. In an industry with an "unattractive" structure, competitors' added values tend to be low, with exceptions arising only in the case of firms that have managed to create competitive advantages for themselves—that is, driven bigger wedges than most of their competitors between buyers' willingness to pay and costs. In more "attractive" industries, a firm may expect to do better than its competitive advantage alone would guarantee, through two mechanisms. First, the added values of individual competitors tend to be larger than their competitive advantages in such industry environments. Second, some industries seem to make it feasible for competitors to engage in what is politely termed "recognition of mutual dependence" and is less politely described as "tacit collusion" (an important determinant of the degree of rivalry as well as a departure from the assumption of unrestricted bargaining).

A PROCESS FOR ANALYSIS

Having reviewed the development of concepts for competitive positioning, we now discuss a process for linking such concepts to strategic planning and action.³² How can managers identify opportunities to raise willingness to pay by more than costs or drive down costs without sacrificing too much willingness to pay? Sheer entrepreneurial insight certainly plays a large role in spotting such arbitrage opportunities. Michael Dell, for example, might see that customers are becoming comfortable with computer technology, realize that retail sales channels add more costs than benefits

for many customers, and act on his insight to start a direct-to-the-customer computer business.³³ Likewise, a company such as Liz Claiborne might perceive a huge pent-up demand for a collection of medium to high-end work clothes for female professionals.³⁴ Dumb luck also plays a role. Engineers searching for a coating material for missiles in the 1950s discovered the lubricant WD-40, the sales of which continued to generate a return on equity between 40 and 50 percent four decades later.

Most strategists believe, however, that smart luck beats dumb luck and that analysis can hone insight. **To analyze competitive advantage, they typically break a firm down into discrete activities or processes and then examine how each contributes to the firm's relative cost position or comparative willingness to pay.**³⁵ The activities undertaken to design, produce, sell, deliver, and service goods are what ultimately incur costs and generate buyer willingness to pay. **Differences across firms in activities—differences in what firms actually do on a day-to-day basis—produce disparities in cost and willingness to pay and hence dictate added value. By analyzing a firm activity by activity, managers can (1) understand why the firm does or does not have added value, (2) spot opportunities to improve its added value, and (3) foresee likely shifts in its added value.**

The starting point of positioning analysis is usually to catalog a business's activities. We can often facilitate the task of grouping the myriad activities that a business performs into a limited number of economically meaningful categories by referring to generic templates for activity analysis, such as the ones reproduced in Exhibits 3.2 and 3.3. Porter's value chain, which distinguishes between primary activities that directly generate a product or service and support activities that make the primary activities possible, is particularly helpful in ensuring that one considers a comprehensive array of activities. Generic templates cannot, however, be used blindly for two reasons. First, not all of the activities they identify will be relevant in any given situation. Second, data often come prepackaged so as to favor a particular way of cataloging activities—unless a major effort to “clean up” such data is deemed necessary.

The rest of the analysis usually proceeds in three steps. First, managers examine the costs associated with each activity, using differences in activities to understand how and why their costs differ from those of competitors. Second, they analyze how each activity generates customer willingness to pay, studying differences in activities to see how and why customers are willing to pay more or less for the goods or services of rivals. Finally, managers consider changes in the firm's activities with the objective of identifying changes that will widen the wedge between costs and willingness to pay.

The following subsections discuss these steps in this order, although it is often necessary to iterate back and forth among them in practice. To illustrate their application, we focus on a simple example: the snack cake market in the western part of Canada.³⁶ Between 1990 and 1995, Little Debbie grew its share of this market from 1 percent to nearly 20 percent. At the same time, Hostess, the maker of such long-time favorites as Twinkies and Devil Dogs, saw its dominant 45 percent share dwindle to 25 percent. An analysis of competitive positioning shows why Little Debbie and Hostess fared so differently and helps suggest a strategy for the latter.

Step 1: Using Activities to Analyze Relative Costs

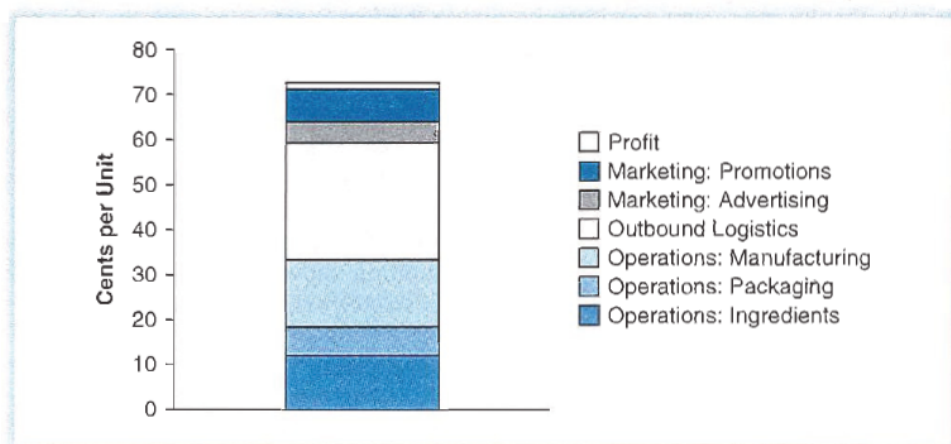
Typically, competitive cost analysis is the starting point for the strategic analysis of competitive advantage. In pure commodity businesses, such as wheat farming, customers refuse to pay a premium for any company's product. In this type of setting, a

low-cost position is the key to added value and competitive advantage. Even in industries that are not pure commodities, however, differences in cost often exert a large influence on differences in profitability.

To begin with our example, in the early 1990s, Hostess's managers struggled to understand why their financial performance was poor and their market share was plummeting. They cataloged the major elements of their value chain and calculated the costs associated with each class of activities. As Exhibit 3.7 shows, although Hostess sold the typical package of snack cakes to retailers for 72¢, raw materials (ingredients and packaging material) accounted for only 18¢ per unit. Operation of automated baking, filling, and packaging production lines (largely depreciation, maintenance, and labor costs) amounted to 15¢. Outbound logistics—delivery of fresh goods directly to convenience stores and supermarkets, and maintenance of shelf space—constituted the largest portion of costs, 26¢. Marketing expenditures on advertising and promotions added another 12¢. Thus, a mere penny remained as profit for Hostess.

After calculating the costs associated with each activity, the managers determined the set of cost drivers associated with each activity, that is, the factors that increase or decrease the cost of an activity. For instance, the Hostess managers realized that the cost of outbound logistics per snack cake fell rapidly as a firm increased its local market share, because delivery costs depended largely on the number of stops that a truck driver had to make. Thus, the larger a firm's market share, the greater the number of snack cakes a driver could deliver per stop. Urban deliveries tended to be more expensive than suburban ones, because city traffic slowed the drivers down. Outbound logistics costs also rose with product variety: the firm's broad product line made it difficult for drivers to restock shelves and remove out-of-date merchandise. Finally, the nature of the product itself affected logistics costs. For instance, snack cakes with more preservatives could be delivered less frequently. Using this information, the managers developed numerical relationships between activity costs and drivers for outbound logistics activities and for the other activities depicted in Exhibit 3.7.

EXHIBIT 3.7 Hostess's Cost Components.



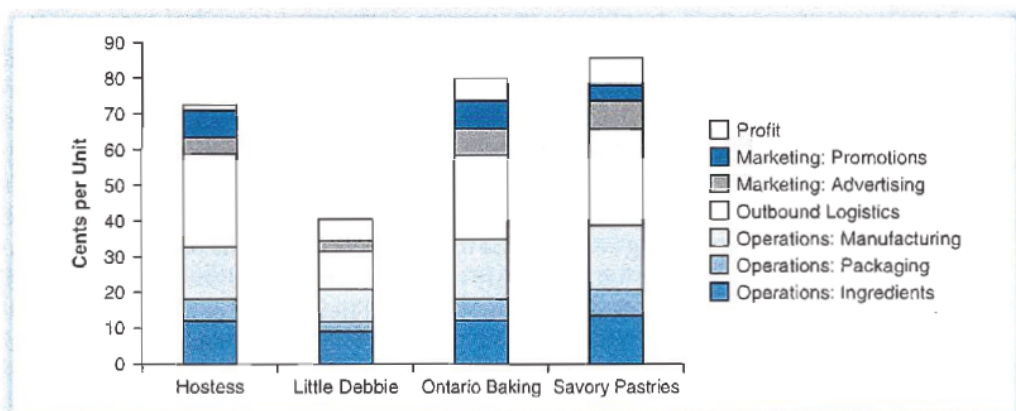
Cost drivers are critical because they allow managers to estimate competitors' cost positions. Although one usually cannot observe a competitor's costs directly, it is often possible to study their drivers. Thus, Hostess could observe Little Debbie's market share, the portion of its sales in urban areas, the breadth of its product line, and the ingredients in its products. Using its own costs and the numerical relationships to cost drivers permitted estimation of the competitor's cost position.

The results of the cost analysis were sobering to Hostess's managers. Because Little Debbie used inexpensive raw materials, purchased in bulk, and tapped national-scale economies, its operations costs totaled 21¢, compared with 33¢ for Hostess. Little Debbie packed its product with preservatives so that deliveries could be made less frequently, kept its product line very simple, and benefited from growing market share. Consequently, its logistics costs per unit were less than half those of Hostess. Also, Little Debbie did not run promotions. Altogether, the managers estimated, a package of Little Debbie snack cakes cost only 34¢ to produce, deliver, and market. Exhibit 3.8 illustrates the results of the cost analysis of Hostess and its major competitors. (The comparison with two other major competitors, Ontario Baking and Savory Pastries, was not so discouraging. Indeed, Hostess had a small cost advantage over each.)

This specific example illustrates a number of general points about relative cost analysis:³⁷

- Managers often examine *actual* costs—rather than opportunity costs—because data on actual costs are concrete and available. Although the symmetric treatment of suppliers and buyers in the formalization of added value is useful—reminding us that competitive advantage can come from better management of supplier relations rather than solely from a focus on downstream customers—supplier opportunity costs and actual costs are usually assumed to track one another closely. Obviously, this assumption should be relaxed when it does not make sense.
- When reviewing a relative cost analysis, it is important to focus on differences in individual activities, not just differences in total cost. Ontario Baking and Savory Pastries, for instance, had similar total costs per unit. The two firms had

EXHIBIT 3.8 Relative Cost Analysis.



different cost structures, however, and, as we will discuss later, these differences reflected distinct competitive positions.

- Good cost analyses typically focus on a subset of a firm's activities. The cost analysis in Exhibit 3.8, for example, does not cover all the activities in the snack cake value chain. Effective cost analyses usually break out in greatest detail and pay the most attention to cost categories that (1) pick up on significant differences across competitors or **strategic options**, (2) correspond to technically separable activities, or (3) are large enough to influence the overall cost position to a significant extent.
- Activities that account for a larger proportion of costs deserve more in-depth treatment in terms of cost drivers. For instance, the snack cake managers assigned several cost drivers to outbound logistics and explored them in depth. They gave less attention to the drivers of advertising costs. The analysis of any cost category should focus on the drivers that have a major impact on it.
- **A cost driver should be modeled only if it is likely to vary across the competitors or in terms of the strategic options that will be considered.** In the snack cake example, manufacturing location influenced wages, rates, and, therefore, operational costs. All of the rivals had plants in western Canada, however, and manufacturing elsewhere was not an option because shipping was costly and goods had to be delivered quickly. Consequently, manufacturing location was not considered as a cost driver.
- Finally, because the analysis of relative costs inevitably involves a large number of assumptions, **sensitivity analysis is crucial.** Sensitivity analysis identifies which assumptions really matter and therefore need to be honed. It also tells the analyst how much confidence to have in the results. Under any reasonable variation of the assumptions, Little Debbie had a substantial cost advantage over Hostess.

A number of references discuss cost drivers in greater detail and suggest specific ways to model them numerically.³⁸ The catalog of potential drivers is long. Many relate to the size of the firm—for example, economies of scale, economies of experience, economies of scope, and capacity utilization. Others relate to differences in firm location, functional policies, timing (e.g., first-mover advantages), institutional factors such as unionization, and government regulations such as tariffs. **Differences in the resources possessed by a firm may also drive differences in activity costs. A farm with more productive soil, for instance, will incur lower fertilization costs.**

A number of pitfalls commonly snare newcomers to cost analysis. Many companies—especially those that produce large quantities of distinct products in a single facility—have grossly inadequate costing systems that must be cleaned up before they can be used as reference points for estimating competitors' costs. As pointed out in courses on management accounting, conventional accounting systems often overemphasize manufacturing costs and allocate overhead and other indirect costs very poorly. As firms move toward selling services and transacting on the basis of knowledge, these outdated systems will make it increasingly more difficult to analyze costs intelligently. Also problematic is a tendency to compare costs as a percentage of sales rather than in absolute dollar terms, which mixes up cost and price differences. Another common but dangerous practice is the mixing together of recurring costs and one-time investments. Analysts sometimes also confuse differences in firms' costs with differences in their product mixes, but one can avoid this problem by comparing the cost positions of

comparable products; for example, one should compare Ford's four-cylinder, midsized family sedan to Toyota's four-cylinder, midsized family sedan rather than some imaginary "average" Ford to an "average" Toyota. Finally, a focus on costs should not crowd out consideration of customer's willingness to pay—the topic of the next section.

Step 2: Using Activities to Analyze Relative Willingness to Pay

The activities of a firm do not just generate costs. They also (one hopes) make customers willing to pay for the firm's product or service. Differences in activities account for differences in willingness to pay and subsequently for differences in added value and profitability. In fact, differences in willingness to pay apparently account for more of the variation in profitability observed among competitors than do disparities in cost levels.³⁹

Virtually any activity in the value chain can affect customers' willingness to pay for a product.⁴⁰ The most obvious effects are those of product design and the manufacturing activities that influence product characteristics—quality, performance, features, aesthetics. For example, consumers pay a premium for New Balance athletic shoes in part because the firm offers durable shoes in hard-to-find sizes. More subtly, a firm can boost willingness to pay through activities associated with sales or delivery—that is, via the ease of purchase, speed of delivery, availability and terms of credit, convenience of the seller, quality of presale advice, and so on. The catalog florist Calyx and Corolla, for instance, can command a premium because it delivers flowers faster and fresher than most of its competitors do.⁴¹ Activities associated with post-sale service or complementary goods—customer training, consulting services, spare parts, product warranties, repair service, compatible products—may also matter. For example, U.S. consumers may hesitate to buy a Fiat automobile because they fear that spare parts and service will be difficult to obtain. Signals conveyed through advertising, packaging, and branding efforts can play a role in determining willingness to pay as well. Nike's advertising and endorsement activities, for instance, affect the premium it commands. Finally, support activities can have a surprisingly large, if indirect, impact on willingness to pay. Thus, the hiring, training, and compensation practices of Nordstrom create a helpful, outgoing sales staff that permits the department store to charge a premium for its clothes.

Ideally, a company would like to have a "willingness-to-pay calculator"—something that indicates how much customers would pay for any combination of activities. For a host of reasons, however, such a calculator usually remains beyond a firm's grasp. In many cases, willingness to pay depends heavily on intangible factors and perceptions that are hard to measure. Moreover, activities can affect willingness to pay in complicated (i.e., nonlinear and nonadditive) ways. And when a business sells to end-users through intermediaries rather than directly, willingness to pay depends on multiple parties.

Lacking a truly accurate calculator, most managers use simplified methods to analyze relative willingness to pay. A typical procedure is as follows. First, managers think carefully about who the *real* buyer is. This determination can be tricky. In the market for snack cakes, for instance, the immediate purchaser is a supermarket or convenience store executive. The ultimate consumer is typically a hungry school-age child. The pivotal decision-maker, however, is probably the parent who chooses among the brands.

Second, managers work to understand what the buyer or buyers want. The snack-cake-buying parent, for example, makes a purchase based on price, brand image, freshness, product variety, and the number of servings per box.⁴² The supermarket or convenience store executive selects a snack cake on the basis of trade margins, turnover, reliability of delivery, consumer recognition, merchandising support, and so forth. Marketing courses discuss ways to flush out such customer needs and desires through formal or informal market research.⁴³ It is important for such research to identify not only what customers *want* but also what they *are willing to pay for*. Moreover, it should reveal the needs that are most important to consumers and determine how they trade them off.

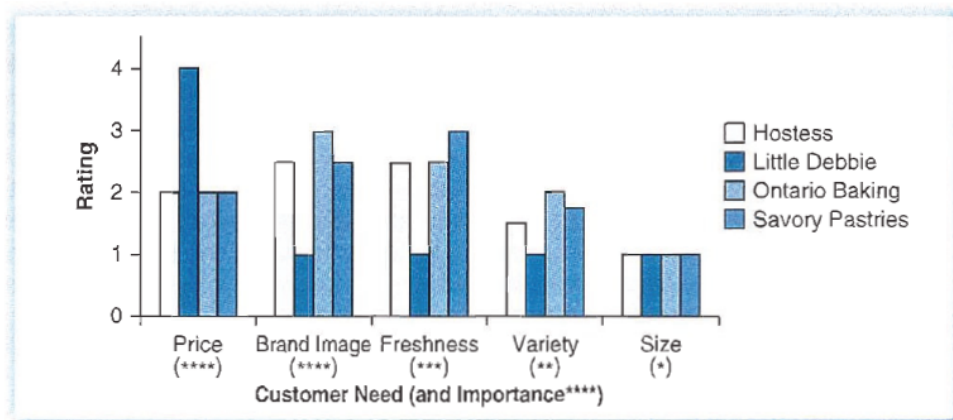
Third, managers assess how successful the firm and its competitors are at fulfilling customer needs. Exhibit 3.9 shows such an analysis for the snack cake market that helps us understand both the statics and the dynamics of the marketplace. Little Debbie stands out on an attribute that customers value highly (low price), while Hostess is not superior on any of the customer needs. This sort of analysis helps explain the large shifts in share observed in the market. Ontario Baking enjoys the best brand image—a position for which it has paid via relatively heavy advertising and promotion. Savory Pastries delivers the freshest product, as reflected in its high manufacturing and raw materials cost. Further analysis (not carried out in the snack cake example) can actually assign dollar values to the various customer needs. For example, it can estimate how much a customer will pay for a product that is one day fresher.

Fourth, managers relate differences in success in meeting customer needs back to company activities. Savory Pastries' high score on the freshness need, for instance, can be tied directly to specific activities regarding procurement and selection of ingredients, manufacturing, and delivery.

At this point, managers should have a refined idea of how activities translate through customer needs, into willingness to pay. They should also understand how activities alter costs. Now they are prepared to take the final step—the analysis of different strategic options. Before moving on to that step, however, we should highlight some guidelines concerning the analysis of willingness to pay.

A major challenge in analyzing willingness to pay is narrowing the long list of customer needs down to a manageable roster. In general, we can ignore needs that

EXHIBIT 3.9 Relative Success in Satisfying Customer Needs.



have little bearing on customer choice. Likewise, needs that are equally well satisfied by all current and contemplated products can usually be neglected. If the group of competing products plays a small role in satisfying a need relative to other products outside the group, the need can often be removed from the list as well.

So far, we have treated all customers as identical. In reality, of course, buyers differ in what they want and how badly they want it. Some customers in a bookstore want novels, while others look for business books. (This type of disparity, in which different customers rank products differently, is known as **horizontal differentiation**.) Among those customers who want J. K. Rowling's new *Harry Potter* novel, some will pay for the more expensive hardback edition, while others are content to wait for the less expensive softcover version. (**Vertical differentiation** arises when customers agree on which product is better—the hardback edition, in this example—but **differ in how much they will pay for the better product**.)

The analysis of willingness to pay becomes trickier, but more interesting, when customers are either horizontally or vertically differentiated. The usual response is **segmentation**: one first finds clumps of customers who share preferences and then analyzes willingness to pay on a segment-by-segment basis. In our experience, firms that identify segments tend to pinpoint between two and twelve clusters of customers. Diversity in customer needs and ease in customizing the firm's product or service typically increase with the number of segments that the analysis considers. Some observers have even argued that companies should move beyond segmentation to embrace mass **customization**.⁴⁴ In this approach, enabled by information and production technologies, companies begin to tailor their products to individual customers. Thus, Blinds To Go receives up to 25,000 custom orders for window blinds and shades per day that it processes within 48 hours. New approaches to customization have enabled the company to build up a business with more than \$100 million in revenues and 20 percent net margins.⁴⁵

Finally, we want to emphasize the limits to analyzing willingness to pay. In some settings, it is possible to quantify willingness to pay quite precisely. For example, when a firm provides an industrial good that saves its customers a well-understood amount of money, it is relatively easy to calculate this amount. Calculations become much more difficult, however, when buyer choice includes a large subjective component, when customer tastes are evolving rapidly, and when the benefits that the customer derives from the product are hard to quantify. A wide range of market research techniques—such as surveys, hedonic pricing, attribute ratings, and conjoint analysis—have been designed to overcome such problems. Nevertheless, we remain wary especially when market research asks people to assess their willingness to pay for new products that they have never seen or for the satisfaction of needs that they themselves may not recognize. Fine market research “proved” that telephone answering machines would sell poorly, for instance.⁴⁶ In some settings, creative insight may have to replace analysis. In all settings, analysis should serve to hone insight and not to displace it.

Step 3: Exploring Different Strategic Options and Making Choices

The final step in the analysis of cost and willingness to pay involves the search for ways to widen the wedge between the two. To this point, the management team has researched how changes in activities will affect added value. The goal now is to find favorable options. Because the generation of options is ultimately a creative act, it is

difficult to lay down many guidelines for it. We can, however, suggest a few patterns from past experience:

1. It is often helpful to distill the essence of what drives each competitor. Little Debbie, for instance, saw that preservatives could substitute for fast delivery. By adding preservatives to its physical product, the company was able to reduce its delivery costs substantially. This approach also reduced customers' willingness to pay, but the reduction was smaller than the corresponding cost savings. Such distillation often suggests new ways to drive wedges between costs and willingness to pay. Savory Pastries, for instance, was tapping a willingness to pay for freshness. The Hostess managers, however, felt that Savory was not exploiting this customer need fully; a product even fresher than what was available from Savory might command a large premium that might serve as the basis for substantial added value.
2. **When considering changes in activities, it is crucial to consider competitor reactions.** In the snack cake example, Hostess's managers felt that Little Debbie would readily launch a price war against any competitor that tried to match its low-cost, low-price position. They were less concerned about an aggressive response from Savory Pastries, whose managers were distracted by an expansion into a different business. Such reactions and, more generally, notions related to sequencing and dynamics are the topics of Chapter 4.
3. Managers often tend to **fixate on a few product characteristics** and think too narrowly about benefits to buyers. **They rarely consider the full range of ways in which all of their activities can create added value.** One way to avoid a narrow focus is to draw out not only one's own value chain but also the value chains of one's customers and suppliers and the linkages between the chains.⁴⁷ Such an exercise can highlight ways to reduce buyers' costs, improve buyers' performance, reduce suppliers' costs, or improve suppliers' performance. Some apparel manufacturers, for instance, have found new ways to satisfy department store buyers that have nothing to do with the physical character of the clothes. By shipping clothes on the proper hangers and in certain containers, for example, these manufacturers can greatly reduce the labor and time required to transfer clothes from the department store's loading dock to the sales floor.
4. **In rapidly developing markets, it is often valuable to pay special attention to leading-edge customers if their demands anticipate the needs of the broader market.** Thus, a number of technology companies release test versions of new services to sophisticated users to shake down software and sense the future needs of the wider market.⁴⁸ Similarly, underserved customer segments often represent a significant opportunity. And the Southwest example reminds us that overserved customers can offer an opportunity as well.
5. More generally, **one of the most potent ways that a firm alters its added value is by adjusting the scope of its operations—that is, changing the range of customers it serves or products it offers within an industry.**⁴⁹ Broad scope in an industry tends to be advantageous when there are significant economies of scale, scope, and learning (including vertical bargaining power based on size), when customers' needs are relatively uniform across market segments, and when it is possible to charge different prices in different segments. Of course, broader is not always

better: there may be diseconomies rather than economies of size, and attempts to serve heterogeneous customers may introduce compromises into a firm's value chain or blur its external or internal message by creating cognitive conflicts in the minds of customers or employees.⁵⁰ And even when broader *is* better, there tend to be a variety of ways in which a firm can expand its reach, some of which (such as licensing, franchises, or strategic alliances) fall short of an outright expansion of scope. Chapters 5 through 7 discuss scope-related issues in detail.

6. We have laid out a process in which a management team develops a comprehensive grasp of how its activities affect costs and willingness to pay, and *then* considers options to widen the wedge between the two. In practice, it is often efficient and effective to reverse this process: to start with a set of options, articulate what each option implies for activities, then analyze the impact of each alternative configuration of activities on the wedge between costs and willingness to pay. By reverse-engineering the analyses they do from the options they face, managers can focus on the analyses that truly matter. Of course, this alternative process works best when managers start with a good grasp of the options available to them.

In general, a firm should search through its business system for, and eliminate, activities that generate costs without creating commensurate willingness to pay. It should also search for inexpensive ways to generate additional willingness to pay, at least among a segment of customers.

THE WHOLE VERSUS THE PARTS

The analysis described in the preceding section focuses on decomposing the firm into parts—that is, discrete activities. In the final step of exploring options, however, the management team must work vigilantly to build a vision of the whole. After all, competitive advantage comes from an *integrated set* of choices about activities. A firm whose choices do not fit together well is unlikely to succeed.

The importance of internal consistency can be visualized, once again, in terms of our metaphor of the business landscape. What particularly complicate the search for high ground—or added value—on this landscape are the interactions among choices: production decisions affect marketing choices, distribution choices need to fit with operations decisions, compensation choices influence a whole range of activities, and so forth. Graphically, the interactions make for a **rugged landscape** characterized by lots of local peaks, each of which depicts a coherent bundle of mutually reinforcing choices.⁵¹

The ruggedness of the business landscape has several vital implications. First, it suggests that incremental analysis and incremental change may not lead an established firm to a new, fundamentally higher position. Rather, the firm may have to consider changing many of its activities in unison to attain a higher peak. To improve its long-run prospects, it may even have to step down and tread through a valley. (Consider the wrenching and far-reaching changes required to turn around IBM, for instance.)

Second, ruggedness implies that more than one internally consistent way to do business often exists within the same industry. Although only a limited number of viable positions are available, more than one high peak usually appears when the interactions among choices are rich. In the retail brokerage business, for instance, both Merrill Lynch and Edward Jones succeeded historically but did so in very different ways. Merrill Lynch operated large offices in major cities, provided access to a

full range of securities, advertised nationally, offered in-house investment vehicles, and served corporate clients as well. Edward Jones operated thousands of one-broker offices in rural and suburban areas, handled only conservative securities, marketed its services through door-to-door sales calls, produced none of its own investment vehicles, and focused almost exclusively on individual investors.⁵²

Third, the landscape metaphor reminds us that the creation of competitive advantage involves *choice*. In occupying one peak, a firm forgoes an alternative position. Relatedly landscape imagery can be used to visualize the role of competition. There is often more added value to occupying one's own separate peak than from piling on to a heavily populated summit. Claiming a distinctive peak may even insulate a new player in an industry from effective retaliation. Consider how Google entered the market for applications such as word processing and spreadsheet software. Had it simply copied Microsoft's approach of selling packaged software to be installed on individual computers, Google would have faced an unrestrained, aggressive response from Microsoft. Instead, Google adopted an altogether different set of choices—scaled a distinct peak: It made word processing and spreadsheet software available online through a Web browser, with documents stored online, with an emphasis on making documents available to collaborators, and without an upfront payment. To follow Google's distinctive approach, Microsoft would have to forego lucrative sales of its own packaged software, Word and Excel. Moreover, it would have to legitimize a Web-based model of computing in which a user's operating system mattered little—an unattractive move for Microsoft, the world's dominant maker of operating systems. Google's focus on a distinctive peak has muted Microsoft's retaliation. Moreover, its approach toward software applications has reinforced its broader strategy of shifting users from a Microsoft-dominated world in which computer processing power resides mostly on personal computers toward a world of "cloud computing," in which users tap into the processing power of centralized servers via Web browsers. Such "new game" strategies and barriers to effective competitive response are discussed further in Chapters 4 and 5.

Finally, the landscape metaphor provides additional perspective on the techniques for industry analyses that were discussed in Chapter 2. It suggests, at least to us, additional emphasis on deaveraging: somewhat less emphasis on analyzing average industry attractiveness—which might be envisioned as the average height of the landscape above sea level—and more emphasis on understanding the industry features that influence the locations of peaks and troughs in the landscape and their evolution over time. The question of how a competitive advantage or disadvantage—related, respectively, to the height of a peak and the depth of a trough—will evolve over time is the focus of the next two chapters of the book, dynamics.

Summary

The analysis in this chapter helps systematize part of the older SWOT (strengths-weaknesses-opportunities-threats) paradigm for strategic planning because strengths and weaknesses often vary substantially, even among direct competitors. As a result, within-industry differences in performance tend to be significant, and businesses that

aim to be very successful must typically position themselves to create competitive advantages within their industries.

Competitive advantage depends on driving a wider wedge between costs and buyers' willingness to pay than one's competitors can. The concept of added value helps integrate considerations of competitive

advantage/disadvantage and industry-level conditions into assessments of the likely profitability of individual businesses. A business has added value when the network of customers, suppliers, and complementors in which it operates is better off with it than without it—that is, when the firm offers something unique and valuable in the marketplace.

To achieve a competitive advantage or a higher added value than its rivals, a business must do things differently from them on a day-to-day basis. These differences in activities, and their effects on relative cost position and relative willingness to pay, can be analyzed in detail and used to generate and assess options for creating competitive advantage.

In addition to decomposing the business into activities, however, its managers must also craft a vision of an integrated whole. Much power can be derived, in particular, from positive, mutually reinforcing

linkages among activities that make the whole more than the sum of its parts.

Finally, we must emphasize the discussions of positioning risk being static instead of dynamic. Some of this risk flows from the terminology itself. It would probably be better to talk about targeting a path for continuous improvement than to discuss settling into a position for all time. Additionally, positioning has a connotation of choosing from a preset, well-specified set of possibilities, whereas coming up with new positions—fundamentally new ways of competing—can have very high payoffs and therefore demands strategic attention. And perhaps most important, while achieving lower costs or delivering greater benefits than competitors can lead to competitive advantage, how is a competitive advantage—or disadvantage—likely to evolve over time? This question occupies center stage in Chapters 4 and 5, which focus, respectively, on competitive dynamics and evolutionary dynamics.

Glossary

added value	focus strategy	strategic options
choice	generic strategies	trade-offs
competitive advantage	horizontal differentiation	unrestricted bargaining
competitive position	internal consistency	value
cost analysis	opportunity costs	value chain
cost drivers	resources	vertical differentiation
customization	rugged landscape	willingness to pay
differentiation	segmentation	
dual competitive advantage	sensitivity analysis	

Notes

1. For further discussion of the definition of competitive advantage, see Steven Postrel, "Competitive Advantage: A Synthesis" (paper presented at the Atlanta Competitive Advantage Conference, Atlanta, June 17, 2004).
2. Dean Neese of Marakon/Trinsum deserves our thanks for helping make these data available.
3. Richard P. Rumelt, "How Much Does Industry Matter?" *Strategic Management Journal* 12, no. 3 (March 1991): 167–185, and Anita M. McGahan and Michael E. Porter, "How Much Does Industry Matter, Really?" *Strategic Management Journal* 18 (July 1997): 15–30.
4. Anita M. McGahan and Michael E. Porter, "The Persistence of Shocks to Profitability: Comparing the Market-Structure and Chicago Views," *Review of Economics and Statistics* 81 (1999): 43–53.
5. Richard E. Caves and Pankaj Ghemawat, "Identifying Mobility Barriers," *Strategic Management Journal* 13 (1992): 1–12; Jan W. Rivkin, "Reconcilable Differences: The

- Relationship Between Industry Conditions and Firm Effects," manuscript, Harvard Business School (1997).
6. See, for instance, Pankaj Ghemawat, "The Risk of Not Investing in a Recession," *Sloan Management Review* 34, no. 2 (Winter 1993): 51–58, reissued with updated commentary in *Sloan Management Review* 50, no. 3 (Spring 2009): 31–38; or David Rhodes and Daniel Stelter, "Seize Advantage in a Downturn," *Harvard Business Review* 87, no. 2 (February 2009): 31–40.
 7. See Michael S. Hunt, "Competition in the Major Home Appliance Industry" (D.B.A. thesis, Harvard University, 1972). Two other dissertations at Harvard—Howard H. Newman, "Strategic Groups and the Structure-Performance Relationship: A Study with Respect to the Chemical Process Industries," and Michael E. Porter, "Retailer Power, Manufacturer Strategy and Performance in Consumer Good Industries"—elaborated and tested the notion of strategic groups. A theoretical foundation for strategic groups was provided by Richard E. Caves and Michael E. Porter, "From Entry Barriers to Mobility Barriers: Conjectural Decisions and Contrived Deterrence to New Competition," *Quarterly Journal of Economics* 91, no. 2 (May 1977): 241–262.
 8. See, for instance, Kenneth J. Hatten and Dan E. Schendel, "Heterogeneity within an Industry: Firm Conduct in the U.S. Brewing Industry, 1952–71," *Journal of Industrial Economics* 26, no. 2 (1977): 97–113.
 9. This conclusion is based on Pankaj Ghemawat's experience working at BCG (Boston Consulting Group) at the beginning of the 1980s.
 10. Michael E. Porter, *Competitive Advantage: Creating and Sustaining Superior Performance* (New York: Free Press, 1985), chap. 2.
 11. For a review of activity-based analysis from the perspective of cost accounting, see Robin Cooper and Robert S. Kaplan, "Profit Priorities from Activity-Based Costing," *Harvard Business Review* 69, no. 5 (May–June 1991): 130–135.
 12. Adapted from Carter F. Bales, P. C. Chatterjee, Donald J. Gogel and Anupam P. Puri, "Competitive Cost Analysis," McKinsey Staff Paper (January 1980), p. 6.
 13. Interestingly, the founders of both Bain and SPA had worked on a BCG study of Texas Instruments that was supposed to have highlighted the problem of shared costs. See Walter Kiechel III, "The Decline of the Experience Curve," *Fortune*, October 5, 1981, p. 139.
 14. Quoted in *ibid.*
 15. The term "differentiated" is often misused. When we say that a firm has differentiated itself, we mean that it has boosted the willingness of customers to pay for its output and can command a price premium. We do not mean simply that the company is different from its competitors. Similarly, it is a common error to say that a company has differentiated itself by charging a lower price than its rivals. A firm's choice of price does not affect how much customers are intrinsically willing to pay for a good—except when price conveys information about product quality.
 16. Interview with Joseph Bower, April 25, 1997.
 17. Interview with Fred Gluck, February 18, 1997.
 18. Michael E. Porter, *Competitive Strategy: Techniques for Analyzing Industries and Competitors* (New York: Free Press, 1980), chap. 2; William K. Hall, "Survival Strategies in a Hostile Environment," *Harvard Business Review* 58, no. 5 (September–October 1980): 75.
 19. Porter, *Competitive Advantage*, p. 33.
 20. Talk by Arnaldo Hax at Massachusetts Institute of Technology, April 29, 1997.
 21. Porter, *Competitive Strategy*, pp. 41–44.
 22. *Ibid.*, chap. 2. Michael E. Porter, "What Is Strategy?" *Harvard Business Review* 74, no. 6 (November–December 1996): 61–78.
 23. Adam M. Brandenburger and Barry J. Nalebuff, *Co-Opetition* (New York: Doubleday, 1996), pp. 127–130.
 24. See, for instance, Pankaj Ghemawat, *Commitment: The Dynamic of Strategy* (New York: Free Press, 1991), chap. 4. For some empirical studies that seem to support this conclusion, see Lynn W. Phillips, Dae R. Chang, and Robert D. Buzzell, "Product Quality, Cost Position and Business Performance: A Test of Some Key Hypotheses," *Journal of Marketing* 47, no. 2 (Spring 1983): 26–43; Danny Miller and Peter H. Friesen, "Porter's (1980) Generic Strategies and Performance: An Empirical Examination with American Data. Part I: Testing Porter," *Organization Studies* 7, no. 1 (1986): 37–55.
 25. W. Chan Kim and Renée Mauborgne, "Blue Ocean Strategy," *Harvard Business Review* 82, no. 10 (October 2004): 76.
 26. "Montreal's 50 Leaders: Guy Laliberté: Shining Star," Montréal Centre-Ville, January, 2007 available at <http://mtlcentreville.v1.myvirtualpaper.com/magazine/2007072701/?page=24>
 27. Adam M. Brandenburger and Harborne W. Stuart, Jr., "Value-Based Business Strategy," *Journal of Economics & Management Strategy* 5, no. 1 (1996): 5–24.

28. Adam M. Brandenburger, Maryellen Costello, and Julia Kou, "Bitter Competition: The Holland Sweetener Co. Vs. Nutrasweet (A)" (Harvard Business School Case no. 794-079, Boston, 1993).
29. This conclusion assumes, of course, that NutraSweet would continue to expand its capacity in line with rapidly growing demand—something that might have appeared quite uncertain to HSC. One limiting feature of added value analysis in its current form is that it doesn't allow for informational complexities of this sort, although it may turn out to be generalizable.
30. In the event, HSC was able, later on, to get paid to stay.
31. Such expedients have variously been referred to as "judo economics" and the "puppy-dog ploy." See Judith R. Gelman and Steven C. Salop, "Judo Economics: Capacity Limitation and Coupon Competition," *Bell Journal of Economics* 14, no. 2 (Autumn 1983): 315–325; Drew Fudenberg and Jean Tirole, "The Fat-Cat Effect, the Puppy-Dog Ploy, and the Lean and Hungry Look," *American Economic Review* 74, no. 2 (May 1984): 361–366.
32. This section draws heavily on ideas first developed in Porter, *Competitive Advantage*, especially chaps. 2–4. Also see Ghemawat, *Commitment*, chap. 4.
33. Das Narayandas and V. Kasturi Rangan, "Dell Computer Corp." (Harvard Business School Case no. 596-058, Boston, 1995).
34. Nicolaj Siggelkow, "Change in the Presence of Fit: The Rise, the Fall, and the Renaissance of Liz Claiborne," *Academy of Management Journal* 44, no. 4 (August 2001): 838–857.
35. Porter, *Competitive Advantage*, chaps. 2–4; Porter, "What Is Strategy?"
36. The authors thank Roger Martin, formerly of Monitor Company, for this example. Details about the companies and other items have been altered substantially to protect proprietary information.
37. See Ghemawat, *Commitment*, chap. 4, for a more extensive list of general guidelines.
38. Porter, *Competitive Advantage*, chap. 3; David Besanko, David Dranove, and Mark Shanley, *The Economics of Strategy* (New York: John Wiley, 1996), chap. 13.
39. Richard E. Caves and Pankaj Ghemawat, "Identifying Mobility Barriers," *Strategic Management Journal* 13, no. 1 (January 1992): 1–12. Of course, this general pattern may not hold up in a particular setting.
40. Porter, *Competitive Advantage*, chap. 4; Besanko, Dranove, and Shanley, *The Economics of Strategy*, chap. 13.
41. Walter J. Salmon and David Wylie, "Calyx & Corolla" (Harvard Business School Case no. 592-035, Boston, 1991).
42. We present "low price" as an attribute that buyers seek. This statement should not be misunderstood as meaning that price determines willingness to pay. Rather, price is included as an attribute in surveys of customer needs so that one can calibrate the willingness of customers to pay a price premium for the other attributes in the survey (such as freshness).
43. See, for instance, Philip Kotler, *Marketing Management: Analysis, Planning, Implementation, and Control*, 8th ed. (Englewood Cliffs, N.J.: Prentice-Hall, 1994).
44. B. Joseph Pine II, *Mass Customization: The New Frontier in Business Competition* (Boston: Harvard Business School Press, 1992).
45. Larry Menor and Ken Mark, "Blinds To Go: Invading the Sunshine State" (Richard Ivey School of Business Case no. 901D04, 2001).
46. Oren Harari, "The Myths of Market Research," *Small Business Reports* 19, no. 7 (July 1994): 48–52.
47. Porter, *Competitive Advantage*.
48. Marco Iansiti and Alan MacCormack, "Developing Products on Internet Time," *Harvard Business Review* 75, no. 5 (September–October 1997): 108–117.
49. Scope has a number of dimensions—horizontal, vertical, and geographic—which were discussed in Chapter 2. The discussion here focuses on horizontal and geographic scope. Vertical scope within an industry, which raises a different set of issues, is discussed at greater length in Chapter 5, in the context of holdup. Horizontal scope that stretches across industries and geographic scope that stretches across national borders are discussed, respectively, in Chapters 6 and 7.
50. Porter, "What Is Strategy?"
51. Daniel A. Levinthal, "Adaptation on Rugged Landscapes," *Management Science* 43, no. 7 (July 1997): 934–950; Jan W. Rivkin, "Imitation of Complex Strategies," *Management Science* 46, no. 6 (June 2000): 824–844. The landscape metaphor is derived from evolutionary biology, especially Stuart A. Kauffman, *The Origins of Order: Self-Organization and Selection in Evolution* (New York: Oxford University Press, 1993).
52. Richard Teitelbaum, "The Wal-Mart of Wall Street," *Fortune*, October 13, 1997, pp. 128–130.